

①

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DL, ML course
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Sem-22
Inferential Statistics

lec-3 2-test: Hypothesis
testing

lec-4 Student's
Distribution

=> Hypothesis testing and
Statistical Analysis

- ① Z test
- ② t test
- ③ CHI square test
- ④ ANOVA test

Z test

The average height of
all the residents in a
city is 168 cm with
 $\sigma = 3.9$

A doctor that mean
to be different

He measured the height of 36 individuals and found the average height to be 169.5 cm.

(a) State the null & alternative hypothesis.

(b) At a 95% confidence level, there is enough evidence to reject the null hypothesis.

(c) For Z-test, we need 2 main conditions

(i) Population std. Deviation
(ii) $n \geq 30$

sol $\mu = 168$ cm

Population mean

Population standard deviation
 $\sigma = 3.9$

$n = 36$

$\bar{x} = 169.5$

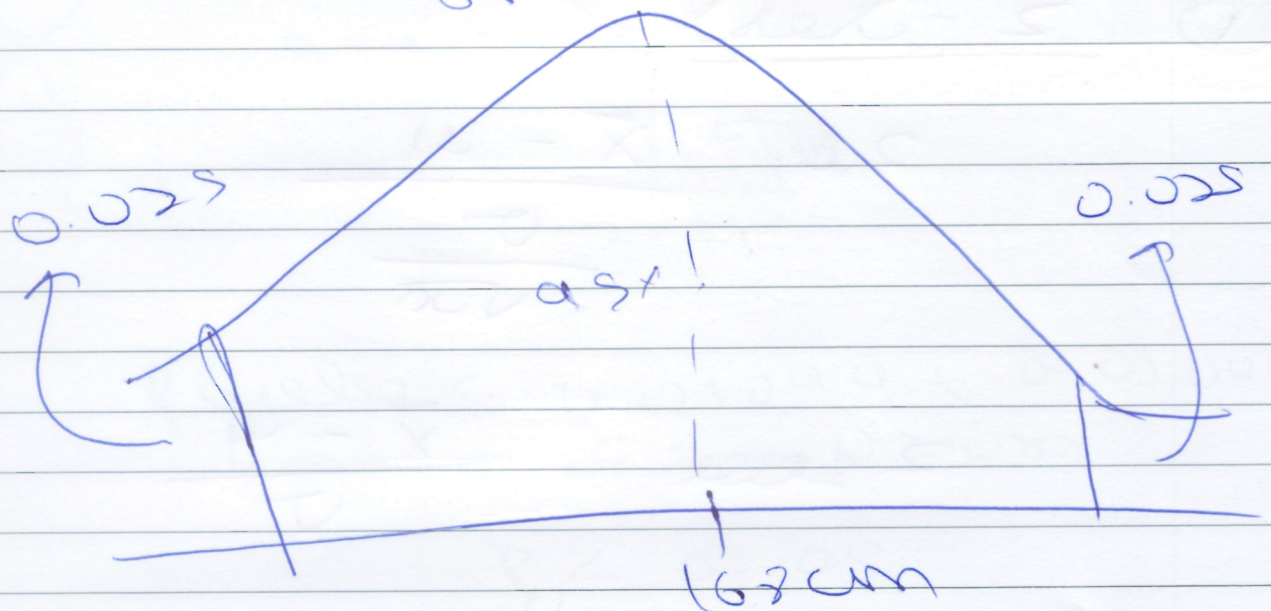
$(\alpha = 0.05)$ (confidence interval)

(3)

Two significance value $\Rightarrow$
 $\alpha = 1 - CF = 1 - 0.45$
 $= 0.05$

① Null Hypothesis
 $H_0: \mu = 168 \text{ cm}$

② Alternative Hypothesis $\mu \neq 168 \text{ cm}$



2 tail test :-

Null Hypothesis
 $\mu = 168 \text{ cm}$

Alternative is that
 $\mu \neq 168 \text{ cm}$

$1 - 0.025 = 0.9750 \Rightarrow$
 Z-score

$$1 - 0.05 = 0.95 \Rightarrow 2\text{-test}$$

$$\frac{1}{0.95} = 1.05$$

2 is less than
-1.96 or greater than
1.96 then
reject the null
Hypothesis.

① 2-test

$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

$$Z_{\text{test}} = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

$$\frac{1095 - 108}{\frac{3.9}{\sqrt{36}}}$$

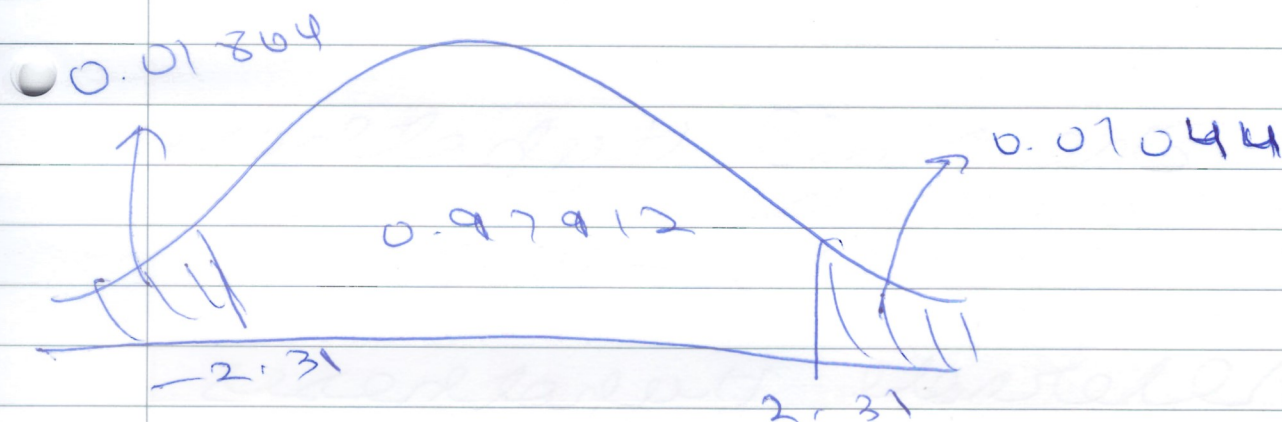
$$\begin{aligned} Z &= \frac{1095 - 108}{\frac{3.9}{\sqrt{36}}} \\ &= \frac{1.5}{0.65} = 2.31 \end{aligned}$$

(5)

therefore, we can see that
 $2.31 > 1.96$

→ reject the
null hypothesis

$$p < 0.05$$



$$p\text{-value} = 0.01044 + 0.01044 \\ = 0.02088$$

$$p < 0.05$$

$$0.02088 < 0.05$$

→ we need to
reject the
null hypothesis

$$\underline{\text{mean} = 16.8 \text{ cm}}$$

⑥
First conclusion, the
Average $\neq 168$ cm

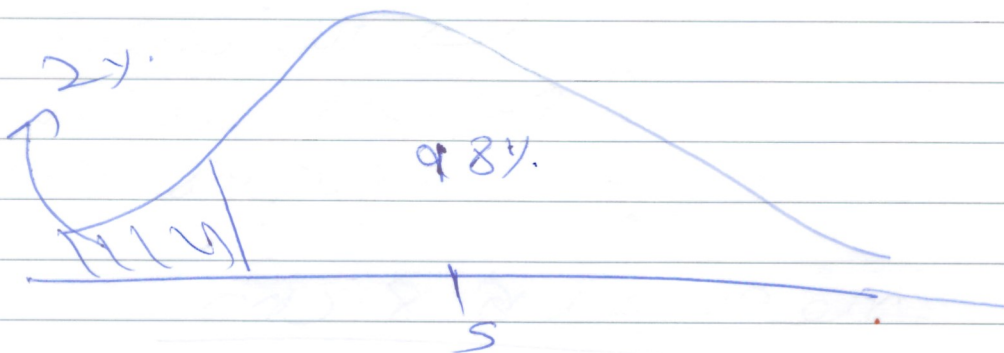
The fuel will
malfunction in less
than 5 years.

(a) One Null Hypothesis
could be
 $H_0: \mu = 5$

Alternate Hypotheses
 $H_1: \mu < 5$

1 tail test

(5) Decision Boundary:-



(2)

(c)

2-test

$$z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

$$= \frac{48 - 5}{\frac{0.5}{\sqrt{40}}}$$

$$= -2.53$$

Area under the curve
with z score = -2.53
= 0.008702

$$p\text{-value} = 0.00870$$

comparison of p-value with

$$\alpha = 0.02$$

→ we accept the
null hypothesis

lec-4

Student's

distribution

• In 2 cases when we perform analysis of variance for 2 - groups.

we require 2 (popul) parameters (mean & standard deviation) as already known.

• Though we perform analysis of variance when we don't know the population std. dev.

Student's t distribution

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

9

s = Sample Standard Deviation

⑩

Degree of Freedom

$$\begin{aligned} \text{dof} &= n - 1 \\ &= 3 - 1 = 2 \end{aligned}$$